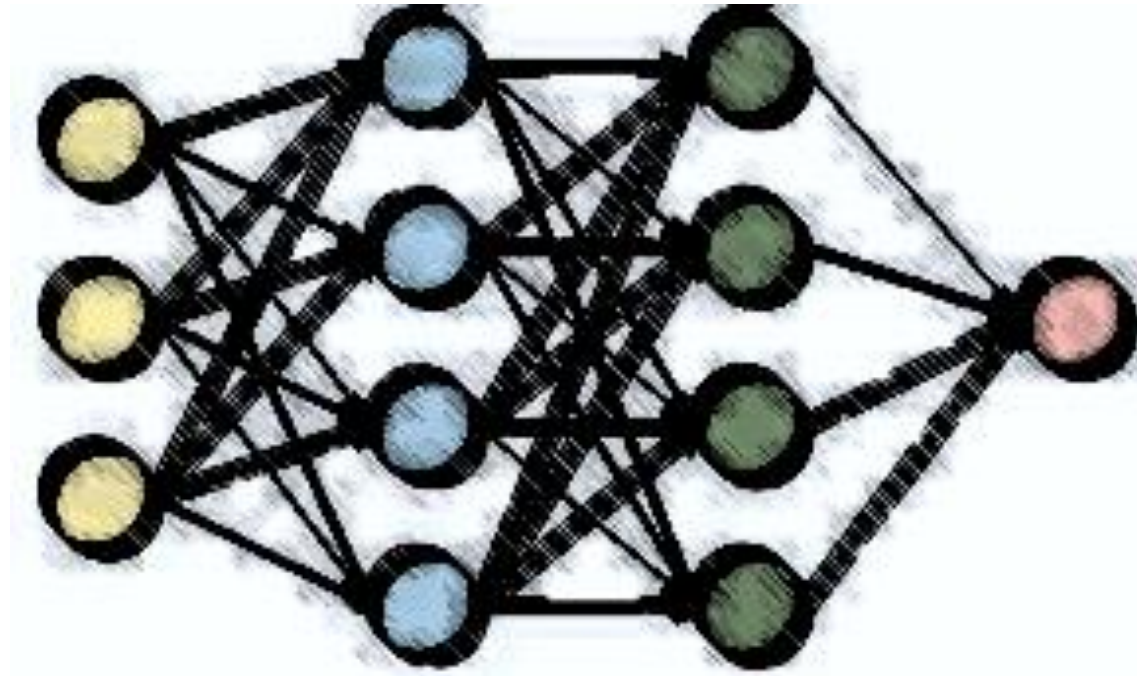
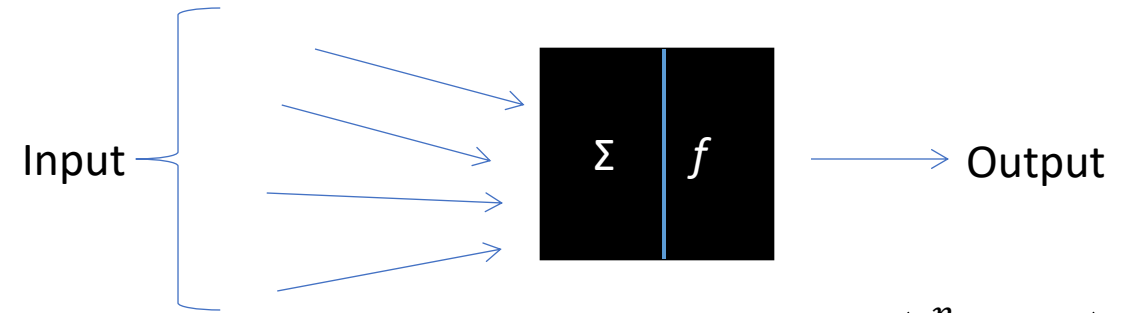
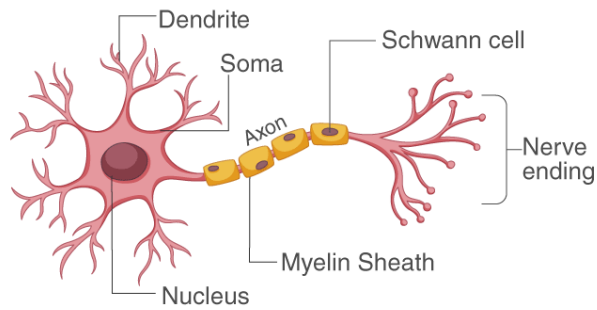


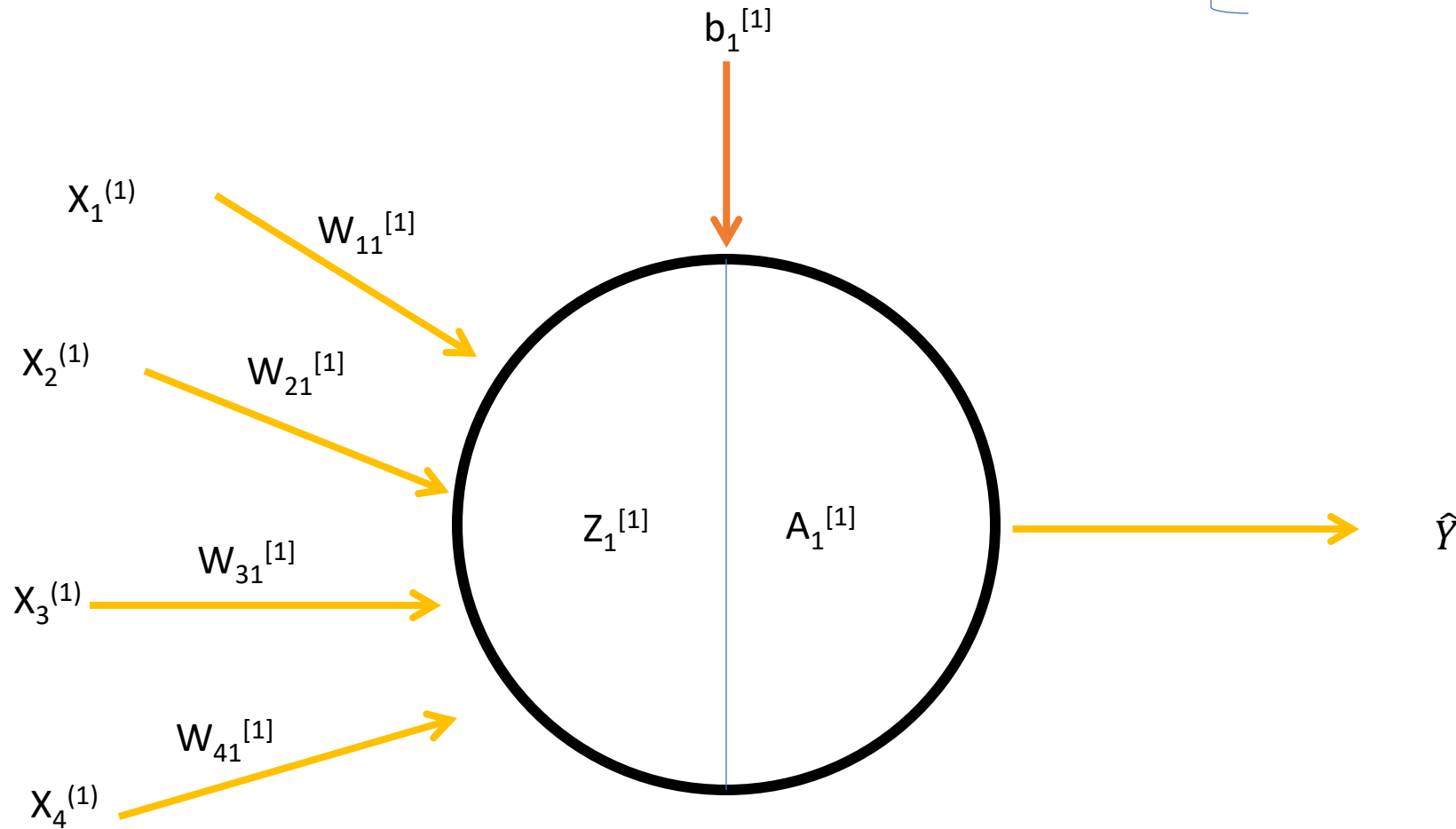
Neural Networks

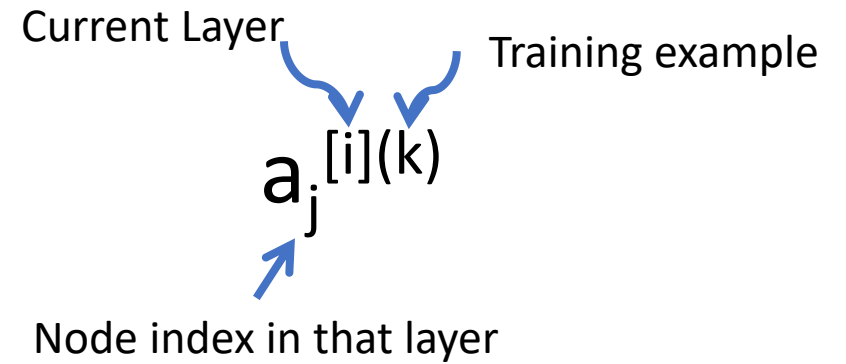
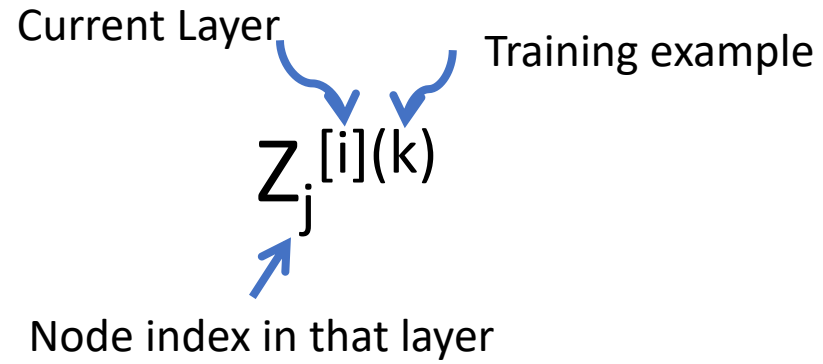
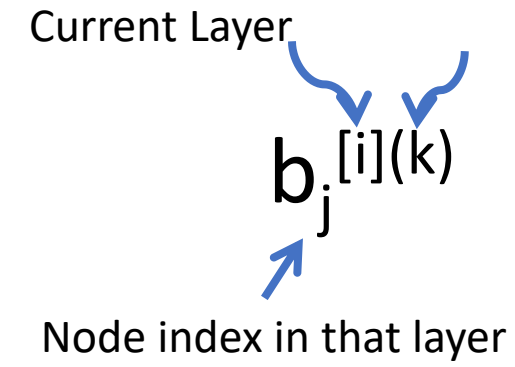
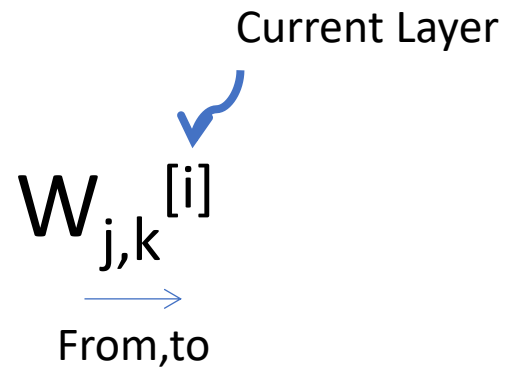


STRUCTURE OF NEURON



$$Output = f\left(\sum_{i=1}^n X_i W_i\right)$$





Single Training Example (i=1) and four input features

Current Layer

One example

$$z_1^{1} = W_{11}^{[1]}X_1^{(1)} + W_{21}^{[1]}X_2^{(1)} + \dots W_{41}^{[1]}X_4^{(1)} + b_1^{[1]} - (1)$$

Node

We can write equation no 1 in Matrix form as:

$$Z_1^{[1]} = \left(W_1^{[1]}\right)^T X + b_1^{[1]} \quad - (2)$$

1 x 1

1 x 4

4 x 1

1 x 1

$$Z_1^{[1]} = [Z_1^{[1]}(1)]$$

$$W_1^{[1]} = \begin{bmatrix} W_{11}^{[1]} \\ W_{21}^{[1]} \\ W_{31}^{[1]} \\ W_{41}^{[1]} \end{bmatrix}$$

$\in \mathbb{R}^{4 \times 1}$

$$X = \begin{bmatrix} X_1^{(1)} \\ X_2^{(1)} \\ X_3^{(1)} \\ X_4^{(1)} \end{bmatrix}$$

$\in \mathbb{R}^{4 \times 1}$

$$Z_1^{[1]} = \left(W_1^{[1]}\right)^T X + b_1^{[1]} \quad - (2)$$

$$a_1^{1} = \sigma \left(Z_1^{[1]}\right) = \hat{Y} \quad - (3)$$

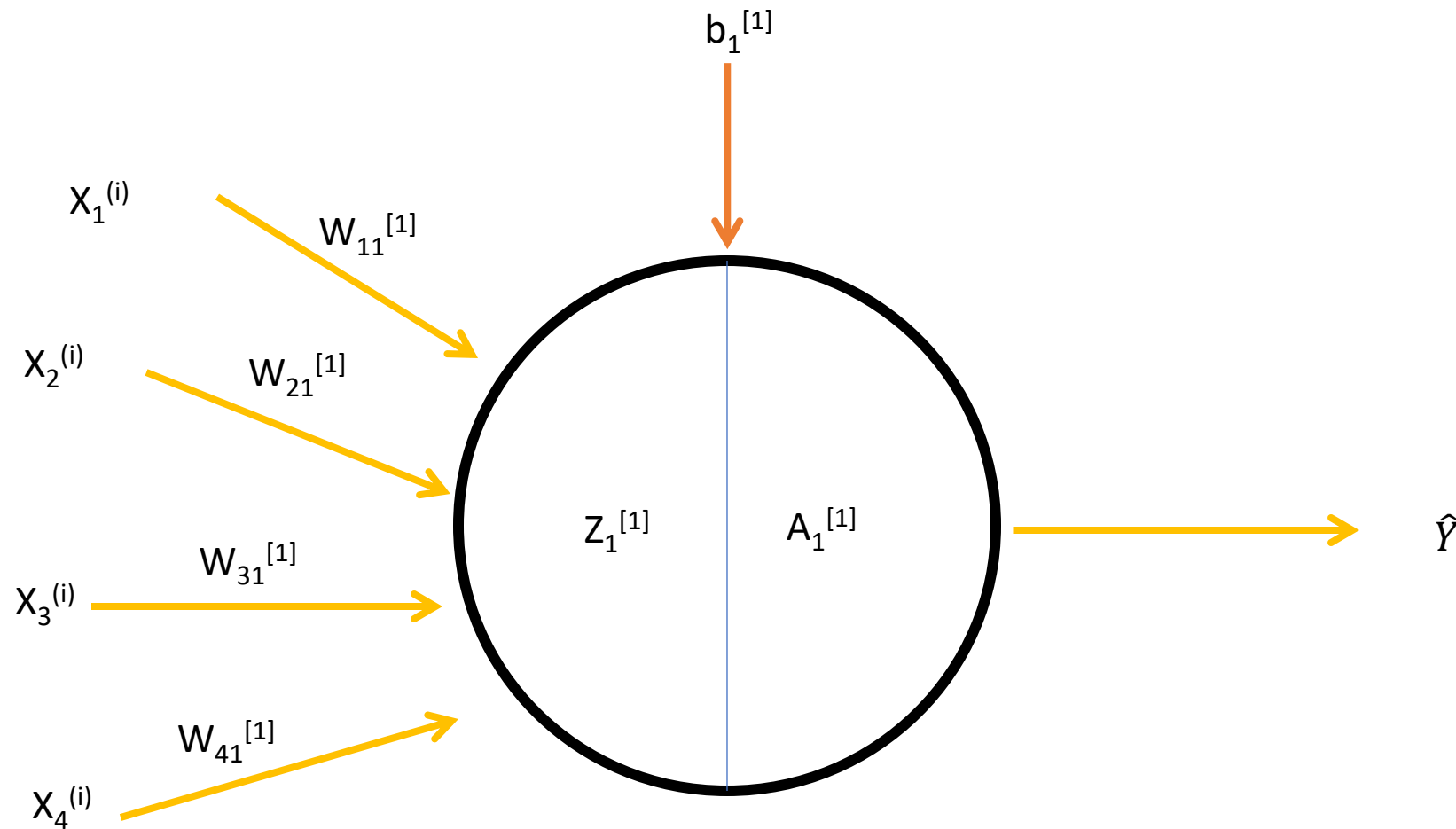
1 x 1



Activation Function

$$A_1^{[1]} = [a_1^{1}] \quad \text{Matrix Form}$$

Multiple Training Examples ($i = 1, 2, 3, 4, 5$) and four features



For n number of features and m training examples we can write X as:

$$X = \begin{bmatrix} X_1^{(1)} & X_1^{(2)} & \dots & X_1^{(m)} \\ \vdots & \vdots & \ddots & \vdots \\ X_n^{(1)} & X_n^{(2)} & \dots & X_n^{(m)} \end{bmatrix}$$

For 4 number of features and 5 training examples we can write X as:

$$X = \begin{bmatrix} X_1^{(1)} & X_1^{(2)} & \dots & X_1^{(5)} \\ \vdots & \vdots & \ddots & \vdots \\ X_4^{(1)} & X_4^{(2)} & \dots & X_4^{(5)} \end{bmatrix}$$

We can write as:

$$Z_1^{1} = W_{11}^{[1]}X_1^{(1)} + W_{21}^{[1]}X_2^{(1)} + \dots W_{41}^{[1]}X_4^{(1)} + b_1^{1}$$

$$Z_1^{[1](2)} = W_{11}^{[1]}X_1^{(2)} + W_{21}^{[1]}X_2^{(2)} + \dots W_{41}^{[1]}X_4^{(2)} + b_1^{[1](2)}$$

.	.
.	.

$$Z_1^{[1](5)} = W_{11}^{[1]}X_1^{(5)} + W_{21}^{[1]}X_2^{(5)} + \dots W_{41}^{[1]}X_4^{(5)} + b_1^{[1](5)}$$

We can combine equations in Matrix form as:

$$Z_1^{1} = \left(W_1^{[1]}\right)^T X^{(1)} + b_1^{1}$$

$$Z_1^{[1](2)} = \left(W_1^{[1]}\right)^T X^{(2)} + b_1^{[1](2)}$$

|
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$$Z_1^{[1](5)} = \left(W_1^{[1]}\right)^T X^{(5)} + b_1^{[1](5)}$$

$$a_1^{1} = \sigma \left(Z_1^{1} \right)$$

$$a_1^{[1](2)} = \sigma \left(Z_1^{[1](2)} \right)$$

$$a_1^{1} = \sigma \left(Z_1^{1} \right)$$

$$a_1^{[1](2)} = \sigma \left(Z_1^{[1](2)} \right)$$

|
.
.

$$a_1^{[1](5)} = \sigma \left(Z_1^{[1](5)} \right)$$

We can write as:

$$Z_1^{[1]} = \left(W_1^{[1]} \right)^T X + b_1^{[1]} \quad - (3)$$

1 x 5

1 x 4

4 x 5

1 x 5

$n = 4$ features and $m = 5$ Training examples

$$Z_1^{[1]} = \left(W_1^{[1]}\right)^T X + b_1^{[1]} \quad - (3)$$

1×5

1×4

4×5

1×5

$$Z_1^{[1]} = [Z_1^{[1])(1)} Z_1^{[1])(2)} \dots Z_1^{[1])(5)}]$$

$$b_1^{[1]} = [b_1^{[1])(1)} b_1^{[1])(2)} \dots b_1^{[1])(5)}]$$

$$A_1^{[1]} = \sigma\left(Z_1^{[1]}\right) = \hat{Y} \quad - (4)$$

1×5

1×5

Generalize Form:

$$Z_1^{[1]} = \left(W_1^{[1]}\right)^T X + b_1^{[1]} \quad - (5)$$

1 x m

1 x n

n x m

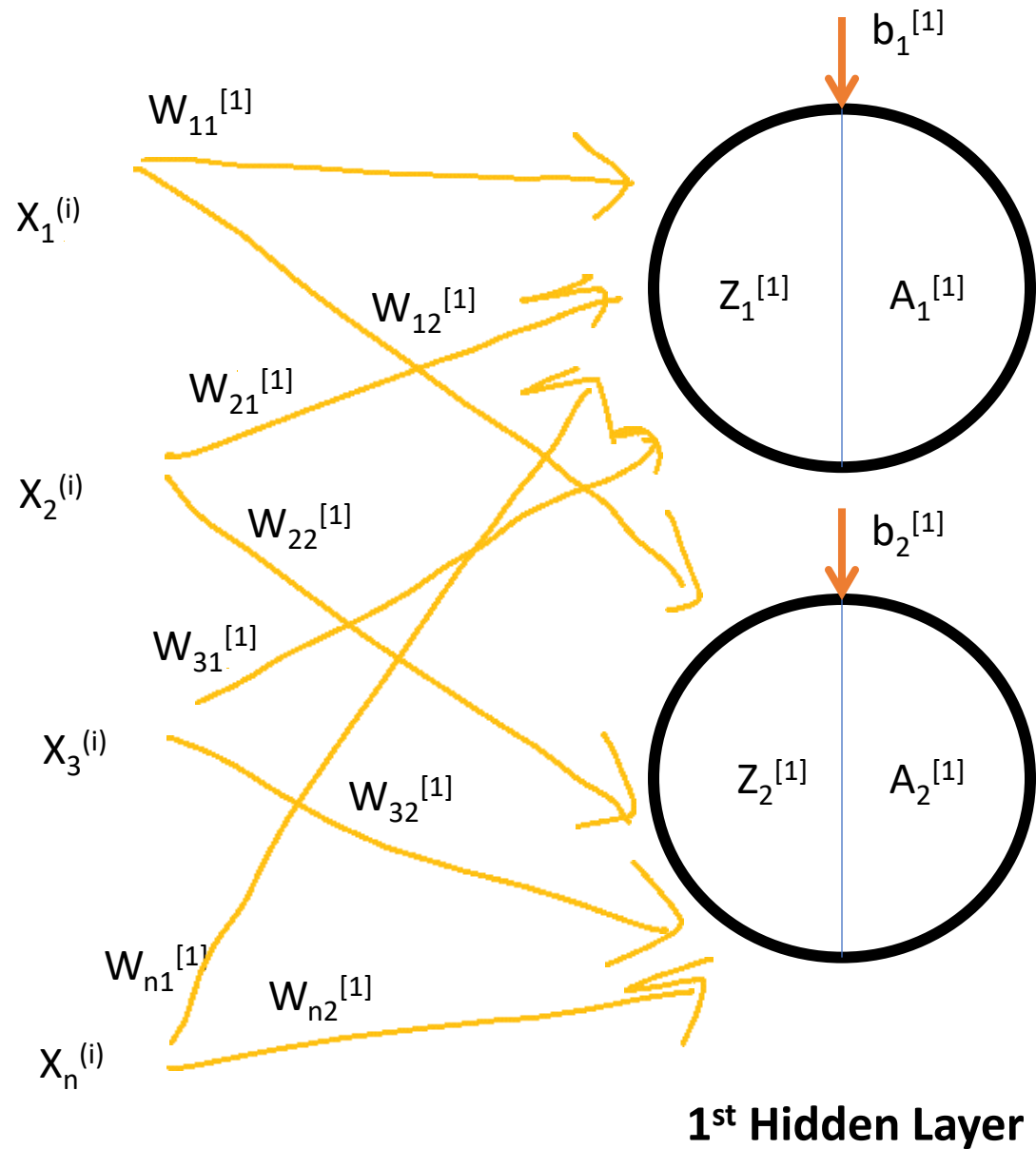
1 x m

This is called as Single Layer Perceptron – Having Single node

Example of Single Neuron

MultiPerceptron

Multi Perceptron



Handwritten notes in blue ink:

$\hat{Y}_1$ $[1]$ $[1]$ $[1]$

$\underline{Z} = WX + B$

$\hat{Y}_2$

$\underline{A} = G(\underline{Z})$

Multi-Layer Perceptron

For Node 2 we can write

$$Z_2^{[1]} = (W_2^{[1]})^T X + b_2^{[1]}$$

Hence, for First Node, Second Node and P nodes

$$Z_1^{[1]} = (W_1^{[1]})^T X + b_1^{[1]}$$

$$Z_2^{[1]} = (W_2^{[1]})^T X + b_2^{[1]}$$

$$Z_p^{[1]} = (W_p^{[1]})^T X + b_p^{[1]}$$

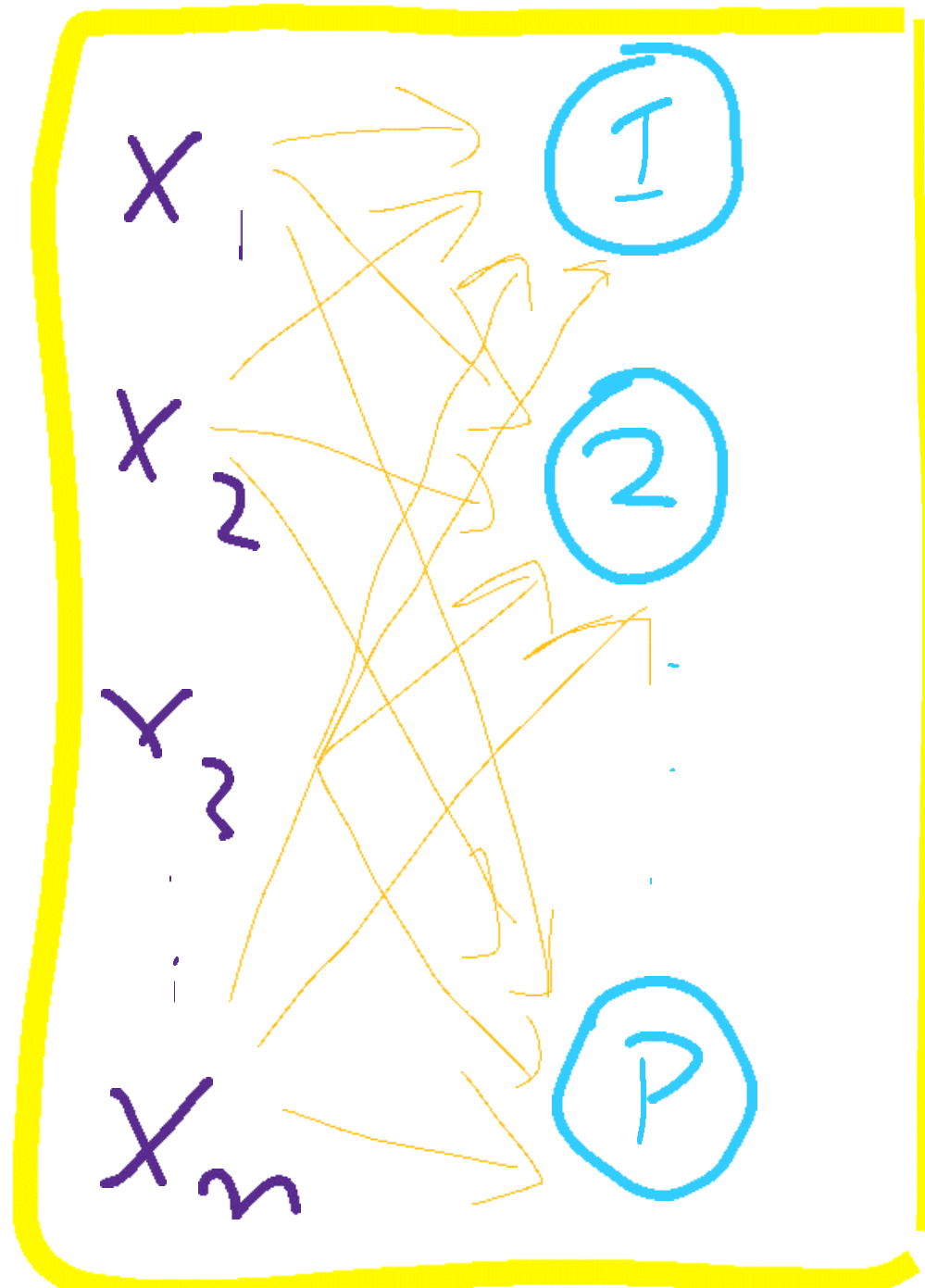
$$Z^{[1]} = (W^{[1]})^T X + B^{[1]} \quad (6)$$

$p \times m$

$p \times n$

$n \times m$

$p \times m$



where,

$$Z^{[1]} = \begin{bmatrix} Z_1^{[1]} \\ Z_2^{[1]} \\ \vdots \\ Z_p^{[1]} \end{bmatrix} = \begin{bmatrix} Z_1^{1} & Z_1^{[1](2)} & \dots & Z_1^{[1](m)} \\ \vdots & \vdots & \ddots & \vdots \\ Z_p^{1} & Z_p^{[1](2)} & \dots & Z_p^{[1](m)} \end{bmatrix}$$

Current layer

Training example

Current node

$$W^{[1]} = \begin{bmatrix} W_1^{[1]} \\ W_2^{[1]} \\ \vdots \\ W_p^{[1]} \end{bmatrix}^T = \begin{bmatrix} W_{11}^{[1]} & \dots & W_{n1}^{[1]} \\ \vdots & \ddots & \vdots \\ W_{1P}^{[1]} & \dots & W_{nP}^{[1]} \end{bmatrix}$$

$$B^{[1]} = \begin{bmatrix} b_1^{[1]} \\ b_2^{[1]} \\ \vdots \\ b_p^{[1]} \end{bmatrix} = \begin{bmatrix} b_1^{1} & b_1^{[1](2)} & \dots & b_1^{[1](m)} \\ \vdots & \vdots & \ddots & \vdots \\ b_p^{1} & b_p^{[1](2)} & \dots & b_p^{[1](m)} \end{bmatrix}$$

And,

$$A^{[1]} = \sigma \left(Z^{[1]} \right) \quad - (7)$$

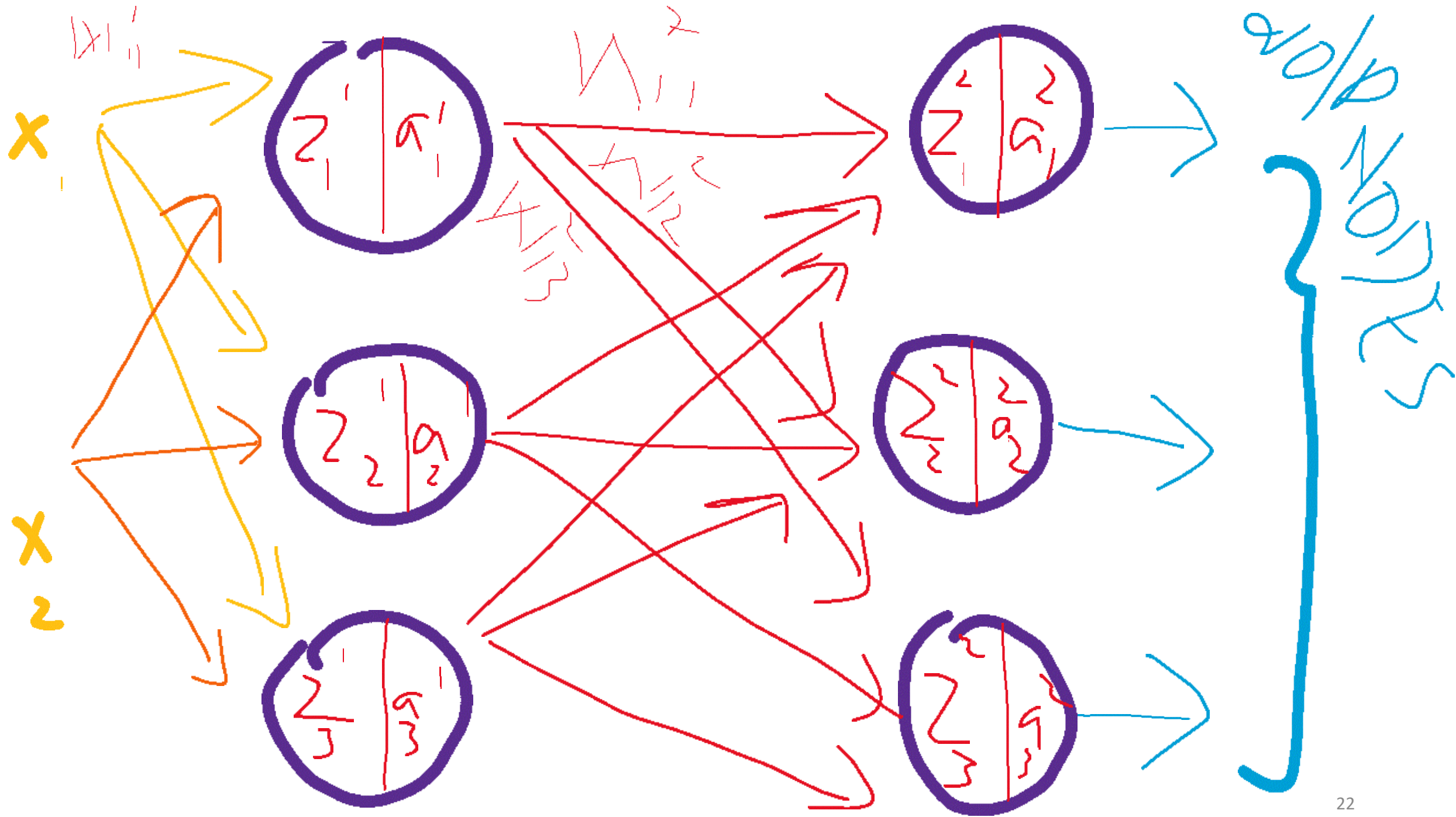
$p \times m$

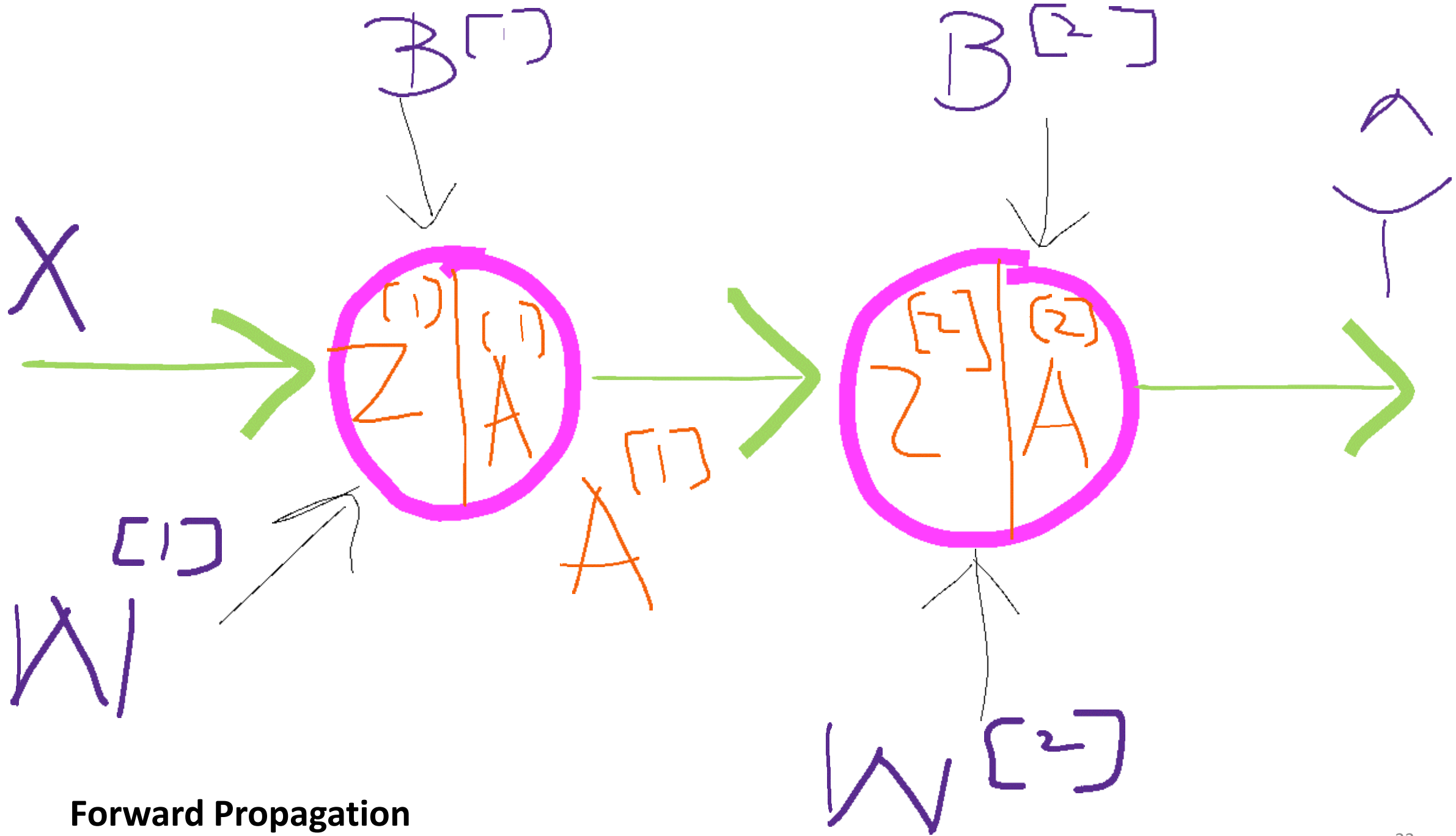
$p \times m$

where,

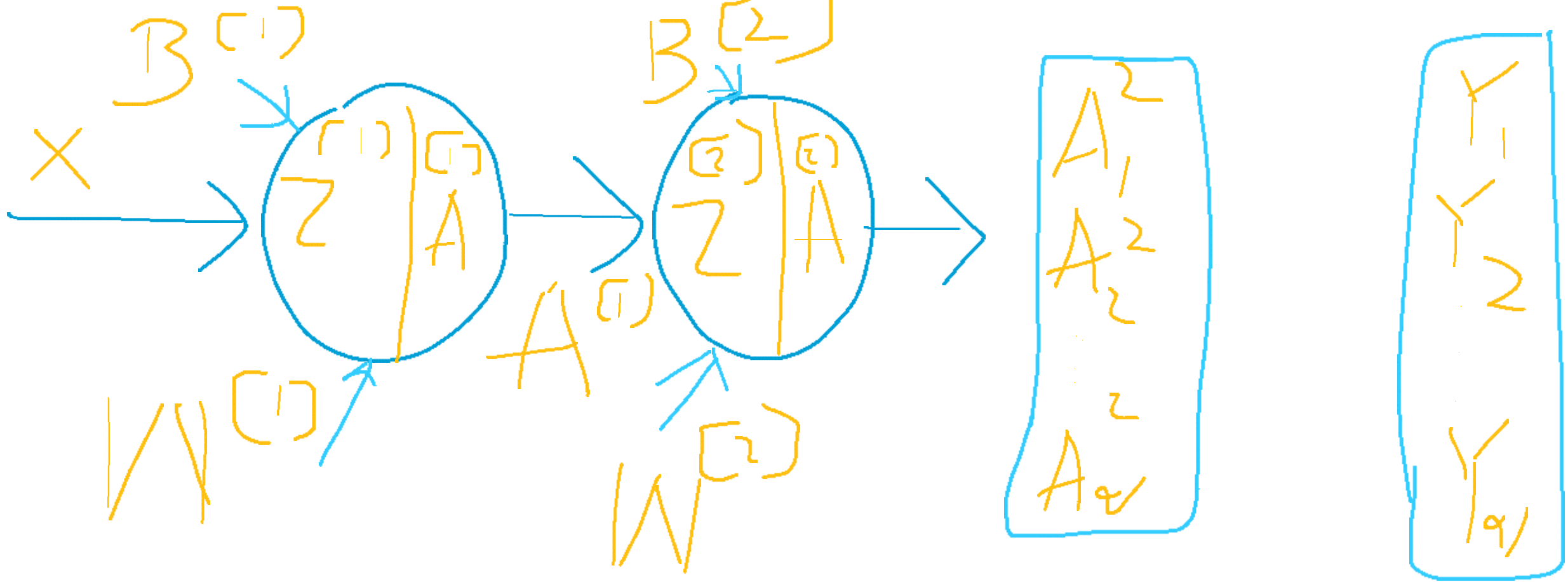
$$A^{[1]} = \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \\ \vdots \\ a_p^{[1]} \end{bmatrix} = \begin{bmatrix} a_1^{1} & a_1^{[1](2)} & \dots & a_1^{[1](m)} \\ \vdots & \vdots & \ddots & \vdots \\ a_p^{1} & a_p^{[1](2)} & \dots & a_p^{[1](m)} \end{bmatrix}$$

Multi-Layer Perceptron





Forward Propagation



$$\text{Loss} = (A - Y)$$

A – Predicted

Y – Actual Labels

Loss

For 1st Layer

$$A^{[1]} = \sigma \left(Z^{[1]} \right) \quad - (8)$$

$p \times m$

$p \times m$

where,

$$Z^{[1]} = \left(W^{[1]} \right)^T X + B^{[1]}$$

$p \times m$

$p \times n$

$n \times m$

$p \times m$

For 2nd Layer

$$A^{[2]} = \sigma \left(Z^{[2]} \right) \quad - (9)$$

$q \times m$

$q \times m$

where,

$$Z^{[2]} = \left(W^{[2]} \right)^T A^{[1]} + B^{[2]}$$

q = output nodes

$q \times m$

$q \times p$

$p \times m$

$q \times m$

Let $q = 2$ i-e two output nodes

In this case: Dimension of $A^{[2]}$ is $2 \times m$

$$A^{[2]} = \begin{bmatrix} A_1^2 \\ A_2^2 \end{bmatrix} = \begin{bmatrix} A_1^2 \\ 1 - A_1^2 \end{bmatrix}$$

Sigmoid output represents probabilities

True output:

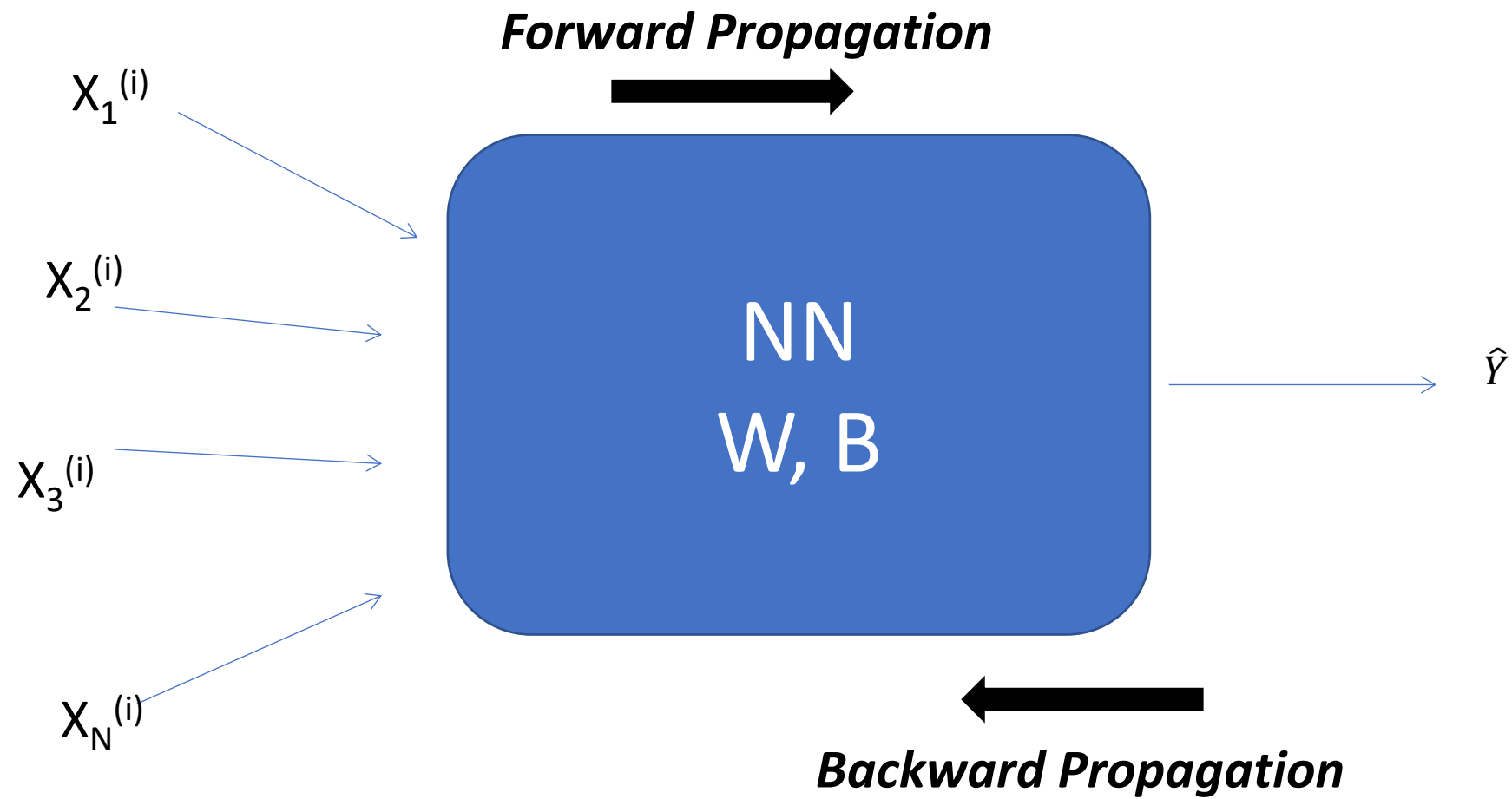
$$Y = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} Y_1 \\ 1 - Y_1 \end{bmatrix}$$

Loss Function:

$$L = - \sum_{i=1}^q Y_i \log A_i^{[2]} \quad \text{Since } q = 2, \text{ binary class}$$

Explore : [Playground.tensorflow.org](https://playground.tensorflow.org)

Back Propagation



Back Propagation (BP)

- We train NN – Forward propagation.
- Backpropagation- Transforming information from output to input.
- We actually tune the weights and bias during Backpropagation.

Updating weights as:

$$w^{new} = w^{old} - \alpha \frac{dl}{dw}$$

- Updating bias as:

$$B^{new} = B^{old} - \alpha \frac{dl}{dB}$$

$$\frac{dl}{dw} \quad \frac{dl}{dB}$$

Both are important to calculate and adjust W and B

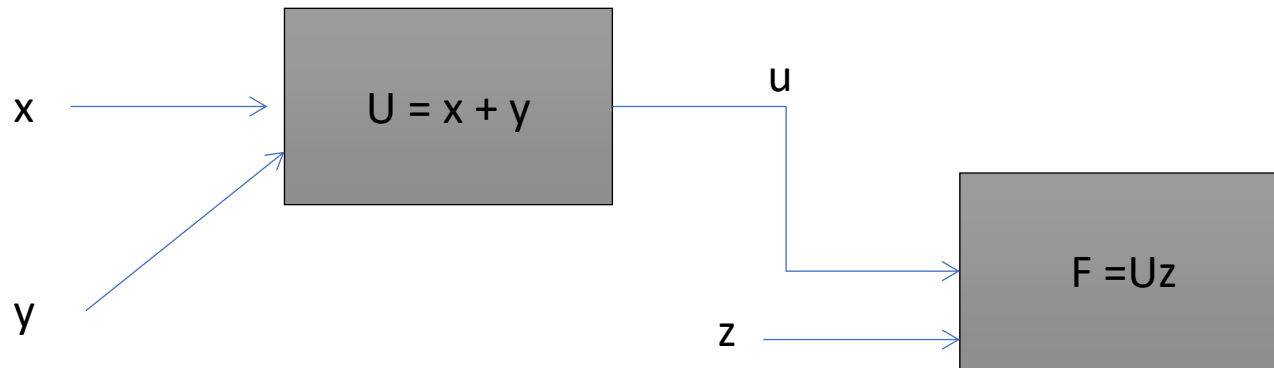
The concept of Computation Graph

- Directed graphs where the nodes correspond to the mathematical operation.

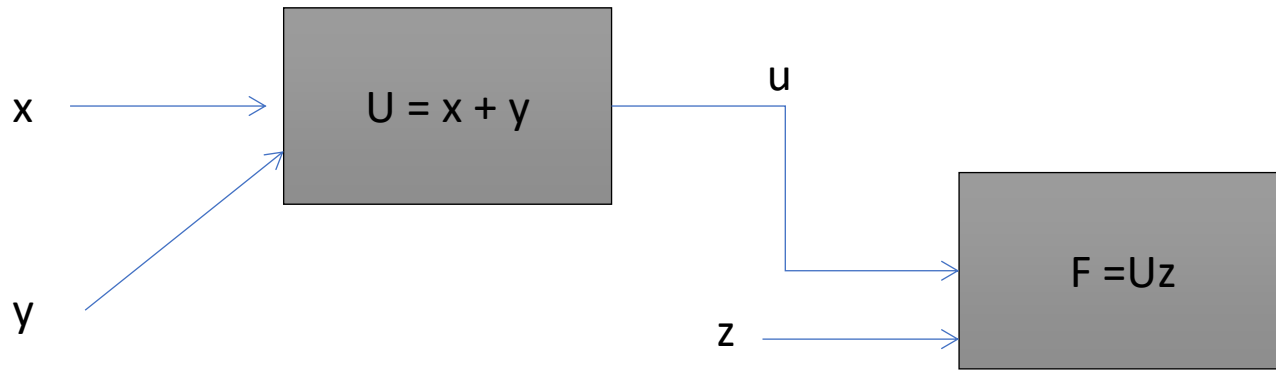
$$F = (x + y)z$$

x , y and z are the forward propagating values.

We can show as,



$X = 1, Y = 2$ and $Z = 3$
 $F = 9$



$$X = 1, Y = 2 \text{ and } Z = 3$$

$$F = 9$$

Now find, $\frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial y}$ and $\frac{\partial F}{\partial z}$

$$U = x + y \text{ and } F = Uz \text{ and } F = xz + yz$$

$$\frac{\partial F}{\partial x} = \left[\frac{\partial F}{\partial U} \right] \left[\frac{\partial U}{\partial x} \right] = z$$

$$\frac{\partial F}{\partial y} = \left[\frac{\partial F}{\partial U} \right] \left[\frac{\partial U}{\partial y} \right] = z$$

$$\frac{\partial F}{\partial z} = x + y = 4$$

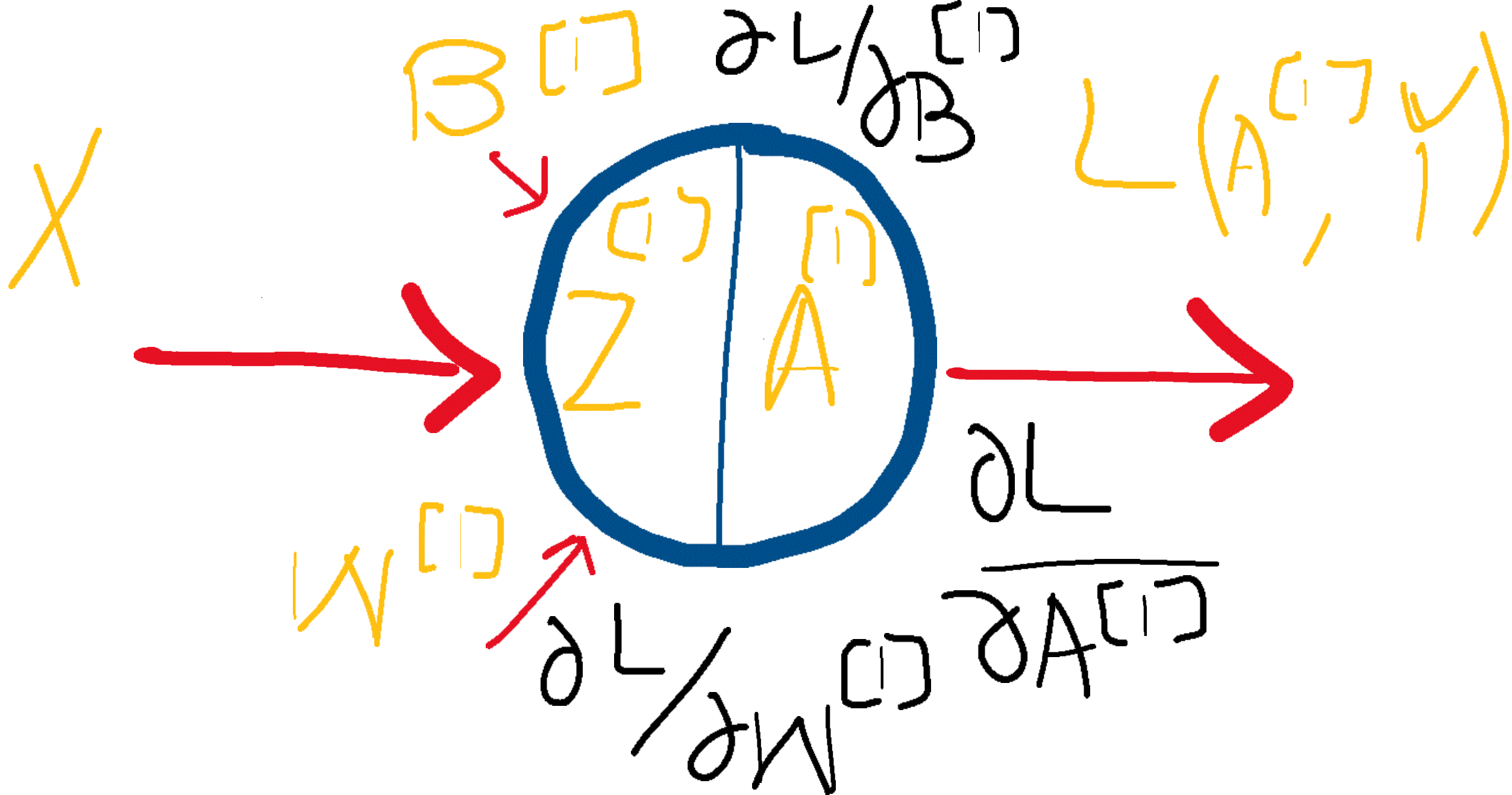
Example 2: $F = 5(XY + Z)$

Self try:

Example 3: $F = 8[(x + y)(y + z) + yz]$

- Self try:





Board Derivation:

